marginal effect of chronic exposure to PM2.5 on first hospital admissions using Medicare data of 11 million participants from seven southeastern states in the United States. A highly refined model was developed to estimate PM2.5 using data from satellite remote sensing, land use and chemical transport models over a 13-year period. The relevance and use of the study for our health impact assessment was enhanced by their sensitivity analysis which examined the risks for concentrations less than 12 μg/m,[3] generally in the same range as in our study area. They reported the following risks of hospitalization for a change of one $\mu g/m^3$ of PM2.5: stroke (5.2%), COPD (7.3%), heart attack (3.4%), pneumonia (10.1%) and heart failure (7.6%).

The findings for cardiovascular disease are supported by at least two other studies on long-term exposure and hospitalization (Miller et al., 2007; Wang et al., 2023). For example, Miller et al. (2007) utilized data from the Women's Health Initiative Observational Study, a prospective cohort study with PM2.5 measured over a six-year period, on average. The outcome examined was first cardiovascular event, which included first occurrence of either myocardial infarction (MI), coronary revascularization, stroke, and death from either coronary heart disease or cerebrovascular disease. Overall, a one $\mu g/m^3$ change in PM2.5 was associated with a 2.4% (95% CI = 0.9, 4.1) increase in a cardiovascular event. When within-city differences in PM2.5 were analyzed, the risks were estimated at 6.4% (95% CI = 2.4, 11.8). Wang et al. (2023) estimated the relationship between annual exposure to PM2.5 and hospitalizations for 10 US states over a 15-year period. In total, almost 2 million MI hospital admissions from 8,106 zip codes were included in this study. The results indicated that when considering only areas where exposures were always less than 12 $\mu g/m^3$ over the study period, the risks were 2.17% (95% CI = 1.79, 2.56) for a one $\mu g/m^3$ change.

These relatively consistent studies provide support for the use of long-term exposures in our assessment. For hospital admissions, therefore, we utilize the findings of Yazdi et al. (2019) since they encompass a suite of health outcomes associated with PM2.5.

3.4.3. *Asthma medication*

Given the high rates of asthma in the cities of Oakland and Richmond (BAAQMD, 2018), it is important to examine the effects of PM2.5 on asthma exacerbation and the need for rescue medication. Rabinovitch et al. (2006) studied a school-based cohort of 73 children followed daily over two winters in Denver, Colorado. Data were collected on peak concentrations of PM2.5 in the morning along with concurrent use of a bronchodilator medication, albuterol. The study reported a 2.6% increase in medication associated with a 12 $\mu g/m^3$ change in peak PM2.5, which was converted to a beta of 0.002 per $\mu g/m^3$. We then utilized the same dispersion patterns as described earlier to determine the increase in medication as a proxy for days with asthma exacerbation.

4. Results

The health impacts were quantified for the three different scenarios described earlier that are based on differing conditions regarding wind speed and direction. Ultimately, the scenarios generated increases in annual average PM2.5 of 0.7, 1.0 and 2.1 $\mu g/m^3$. Table 2 summarizes the estimated health impacts for the 262,000 people who are exposed to an increase in PM2.5 from the passing coal trains. Although this is a relatively small population, it is a microcosm of what likely occurs to millions of urban dwellers throughout the world as uncovered coal trains traverse the globe delivering coal to power plants and export terminals.

Focusing on the most severe outcomes for this study population, across the three exposure scenarios the annual mortality impacts of coal trains range from 2 to 6 deaths per year in the basic model and from 5 to 15 in the race-adjusted model. Hospital admissions for cardiovascular disease and pneumonia range from 9 to 28, and 6 to 17, respectively, over the three scenarios. For asthma, over the three scenarios, new cases range from 7 to 22, while asthma symptoms range from 19,000 to 58,000 days attributable to coal train transit. With asthmatics making up 6.4% of the 6 to 17 age group, this means an average of about 10 symptom days per year per asthmatic. These estimates amount to increases of between one and six percent over baseline levels.

Based on the middle scenario of a one $\mu g/m^3$ change in annual average, there is an estimated increase per year of three (basic model) to seven deaths (race-adjusted model), five hospital admissions for chronic lung disease, nine for congestive heart failure, eight for pneumonia, six for stroke and 14 for all cardiovascular disease. The analysis also indicates about 11 new cases of asthma and 27,000 additional asthma symptoms requiring medication. Among the less severe outcomes, 1,300 additional cases of hay fever/rhinitis, 3,300 days with some restrictions in activity and 560 additional days of work loss are estimated.

Table 3 summarizes the analysis of relative PM2.5 increments by race/ethnicity. The results indicate that Hispanic and Black residents in the study area have a 41% and 29% higher level of exposure to PM2.5, respectively, relative to White residents.

5. Discussion

Our study is the first to estimate the health impacts of uncovered passing coal trains. The central finding is that exposures to PM2.5 from these trains are associated with significant mortality and morbidity.

Table 2
Estimated Health Effects and Associated Confidence Intervals due to passage of coal trains.

Endpoint	Estimated Cases for Alternative Peak Annual Average Increments (μg/m3)		
	0.7	1.0	2.1
Mortality, WHO Basic	2.0 (1.6, 2.4)	2.8 (2.4, 3.2)	5.9 (4.8, 6.8)
Mortality, WHO <12 $\mu g/m^3$	2.8 (2.3, 3.2)	3.9 (3.2, 4.6)	8.2 (6.7, 9.6)
Mortality, Race-Adjusted	5.1 (4.2, 6.0)	7.3 (6.0, 8.5)	15.2 (12.5, 17.8)
HA Chronic Lung Disease[a]	3.5 (3.4, 3.7)	5 (4.8–5.2)	10.4 (10–10.8)
HA Congestive	6.4 (6.1–6.6)	9.1 (8.7–9.4)	18.8 (18–19.5)
HA Pneumonia	5.9 (5.6–6.1)	8.3 (8–8.6)	17.1 (16.5–17.8)
HA Stroke	4.1 (3.6–4.7)	5.9 (5.1–6.6)	12.2 (10.5–13.8)
HA All Cardiovascular	9.4 (8.3–10.5)	13.4 (11.8–15)	28 (24.6–31.1)
Acute MI, Nonfatal	1.1 (0.2–1.8)	1.5 (0.3–2.6)	3.2 (0.7, 5.5)
Asthma Symptom Days[b]	19.3 (6.7, 31.0)	27.0 (19.5, 43.3)	57.6 (201, 91.9)
New Asthma (age 0–4)	4.4 (4.3–4.6)	6.3 (6.1–6.6)	13.2 (12.6–13.7)
New Asthma (age 5–17)	3.1 (2.9–3.2)	4.4 (4.2–4.5)	9 (8.7–9.4)
Hay Fever/Rhinitis	45 (11–78)	64 (15–111)	134 (33–231)
MRAD	2.3 (1.9–2.7)	3.2 (2.7–3.9)	6.9 (5.6–8.1)
Work Loss Days	395 (333–455)	564 (476–650)	1184 (998–1363)

Note: HA = Hospital admissions; HF = Heart Failure; RAD = Restricted Activity Days.
[a] Not including asthma
[b] In thousands

Table 3
Study Population and Associated Increment in annual average PM2.5

	Hispanic	Asian	Black	NatAmer	White	Total
Population	90686	44649	57334	845	68525	262039
% Population	34.6%	17.0%	21.9%	0.3%	26.2%	100.0%
Annual Average PM2.5 Increment	0.82	0.62	0.75	0.67	0.58	0.71
Relative PM2.5 Levels	1.15	0.88	1.07	0.95	0.81	1.00

While the absolute number of cases is relatively small due to the size of the catchment area, under certain scenarios they amount to increases of one to six percent over baseline. Further, these findings provide a microcosm of what might be expected globally given the multiple millions that are exposed by the common traverse of coal trains. This study based its parameters on a specific – though generalizable – coal export proposal and concludes that the anticipated number of weekly mile-long trains that would be delivering coal to the proposed terminal will result in chronic exposure to elevated PM2.5. The quantification of subsequent health impacts is noteworthy for finding previously unrecognized and severe impacts such as death and hospitalization and for including the addition of several morbidity outcomes associated with long-term exposures.

For example, in the worst-case scenario of a 2.1 $\mu g/m^3$ increase near the rail line, premature mortality would increase by 1.3% while hospital admissions for chronic lung disease, pneumonia and cardiovascular disease would increase by 4.7%, 6.2% and 2.2%, respectively. In addition, the number of new cases of asthma and the number of days of asthma exacerbation would increase by about 3.2%. These percent increases are reduced by half and two-thirds in the 1.0 $\mu g/m^3$ and 0.7 $\mu g/m^3$ scenarios, respectively.

While the long-term exposure impacts are noteworthy, it is important to note that there is epidemiological evidence of additional impacts after exposures as short as one or more hours. This is relevant due to the high peaks (5-min average) of PM2.5 observed at the rail line, The subacute impacts of PM2.5 on asthma exacerbation was included in above estimates. However, there are also other studies indicating that exposures as short as 1 h (or a few hours) can increase the risk of outcomes such as acute MI, arrhythmia, hospitalization and emergency department visits for cardiovascular and respiratory disease and ambulance calls (Peters et al., 2004; Kim et al., 2015; Chen et al., 2022; Wu et al., 2020).

For several reasons, the impacts in this assessment are likely underestimated. First, several important health impacts either are not included in BenMAP or were not estimated due to lack of sample size in the study population. These include adverse birth outcomes (i.e., prematurity, low birthweight), neurological impacts on infants and cognitive impacts on adults (i.e., depression, Parkinson's and Alzheimer's diseases) and lung and other organ cancers. Second, we have not estimated health impacts associated with exposures to ultrafine or coarse particles from the passing trains.

Third, the baseline rates of mortality, heart disease and asthma are higher in the study area than those used in the BenMAP software, which are county-level data. For example, data from the Alameda County Public Health Department shows that in West Oakland (the site where the proposed terminal would be located and where trains would pass through), the rates of asthma emergency room visits and hospitalizations, heart disease and stroke mortality, and cancer rates are all about twice the county average. The Department also confirms that the Oakland population living within one mile of rail lines is markedly different demographically than that living farther away, with a higher percentage of people of color, children, and adolescents, and people living in poverty (ACPHD, 2016).

Likewise, for the study population located in Richmond, several of the census tracts abutting the rail line have mortality rates 10%–50% higher than the county average. The higher risk and baseline levels for the study population relative to county estimates would obviously result in higher quantitative estimates of the health impacts. Finally, the estimates do not include the potential PM2.5 exposures and impacts of loading and unloading at the proposed terminal itself.

There are also several limitations and uncertainties in the assessment. First, the original estimates of the impact of passing uncovered coal cars are based on only three studies, although the results of the two larger studies are very consistent (Ostro et al. 2023a, 2023b; Jaffe et al., 2015). Second, due to lack of available studies that specifically measured the dispersion of PM2.5 from trains, we relied on a single study for the dispersion model. However, the actual measurement of PM2.5 concentrations in that study often provides a better measure than dispersion models based on emissions factors. Interestingly, in this case, our findings for the increments are similar to those provided by the dispersion model for coal trains developed by Grey (2017). Third is the assumption of equal toxicity amongst the components of PM2.5, a common assumption in most impact assessments including the Global Burden of Disease (Fuller et al., 2022).

Taken together, our estimates provide a reasonable assessment of the likely adverse health effects of uncovered coal cars traversing through urban areas throughout the world. They indicate that coal's impacts, including mortality, extend beyond coal combustion and mining to its rail conveyance and export. Given that the three largest users of coal are India, China, and the U.S., and that over 80 countries use coal, this study confirms that large populations are impacted by coal trains, making this practice a heretofore insufficiently studied or regulated public health concern of global proportions.

Unfortunately, there are limited possibilities for mitigation of the particle emissions of passing coal trains. Traditionally, chemical surfactants have been applied to the coal piles, but they tend to degrade over time and space. Coal car covers are not commercially available and if used could increase the risk of spontaneous combustion since coal is exothermic. In addition, even if available, the covers would render the coal cars heavier, leading to the risk of impairment of the train tracks and potential derailment.

That the young and old, low income, and communities of color are disproportionately burdened by heightened exposure and vulnerability to rail transport of coal as well as to the climate impacts of its combustion indicate that the matter of coal train activity also constitutes a matter of environmental justice.

There was no completing interests

This work was supported in part by the California Air Resources Board Community Air Monitoring Grant Program (Grant G19-CAGP-17). The funding source had no involvement in any aspect of the study itself. Additional support was provided by the UC Davis Environmental Health Science Center Core P30 ES023513 Pilot Project #87C49.

CRediT authorship contribution statement

Bart Ostro: Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yuanyuan Fang:** Writing – review & editing, Formal analysis, Data curation. **Marc Carreras Sospedra:** Writing – review & editing, Methodology, Formal analysis, Data curation. **Heather Kuiper:** Writing – original draft, Project administration, Funding acquisition, Data curation. **Keita Ebisu:** Writing – review &

editing, Methodology, Data curation. **Nicholas Spada:** Writing – review & editing, Project administration, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

ACPHD, 2016. Alameda County Public Health Department, Community Assessment, Planning, and Evaluation.

Akaoka, K., McKendry, I., Saxton, J., et al., 2017. Impact of coal-carrying trains on particulate matter concentrations in South Delta, British Columbia, Canada. Environ. Pollut. 223, 376–383. https://doi.org/10.1016/j.envpol.2017.01.034.

BAAQMD, 2018. San Francisco Bay Area Community Health Protection Program: Improving neighborhood air quality final submittal: Public Process for determination of recommended communities. Bay Area Air Quality Management District,. August 1, 1-22.

Chen, J, Hoek, G, 2020. Long-term exposure to PM and all-cause and cause-specific mortality: A systematic review and meta-analysis. Environ Int 143, 105974. https://doi.org/10.1016/j.envint.2020.105974.

Chen, R., Jiang, Y., Hu, J., et al., 2022. Hourly air pollutants and acute coronary syndrome onset in 1.29 million patients. Circulation 145, 1749–1760.

Chen, J., Braun, D., Christidis, T., et al., 2023. Long-Term Exposure to Low-Level PM 2.5 and Mortality: Investigation of Heterogeneity by Harmonizing Analyses in Large Cohort Studies in Canada, United States, and Europe. Environ. Health Perspect. 131 (12), 127003.

Colmer, J., Hardman, I., Shimshack, J., et al., 2020. Disparities in PM2.5 air pollution in the United States. Science 369 (6503), 575–578. https://doi.org/10.1126/science.aaz9353.

Crouse, D., Erickson, A., Christidis, T., et al., 2020. Evaluating the sensitivity of pm2.5–mortality associations to the spatial and temporal scale of exposure assessment. Epidemiology 31 (2), 168–176. https://doi.org/10.1097/EDE.0000000000001136.

Di, Q., Wang, Y., Zanobetti, A., et al., 2017. Air pollution and mortality in the Medicare population. N. Engl. J. Med. 376 (26), 2513–2522.

Energy Institute, 2023. Statistical review of world Energy 2022. https://www.energyinst.org/statistical-review/home.

Freightwaves, 2019. Commentary: Is the Death of Rail-Hauled Coal True, or Premature? https://www.freightwaves.com/news/is-the-death-of-rail-hauled-coal-true-or-premature#:~:text=It%20comprises%20about%2014%20percent,rail%20are%20up%20and%20down.

Fuller, R., Landrigan, P.J., Balakrishnan, et al., 2022. Pollution and health: a progress update. Lancet Planet. Health 6 (6), e535–e547. https://doi.org/10.1016/S2542-5196(22)00090-0.

Grey, H., 2017. Modeling PM air quality impacts of the proposed OBOT facility. Gray Sky Solutions, San Rafael. CA, October 6, 1–35.

Henneman, L., Choirat, C., Dedoussi, I., et al., 2023. Mortality risk from United States coal electricity generation. Science 382, 941–946. https://www.science.org/doi/epdf/10.1126/science.adf4915.

International Energy Agency, 2023. https://www.iea.org/energy-system/fossil-fuels/coal.

Jaffe, D., Putz, J., Hof, G., et al., 2015. Diesel particulate matter and coal dust from trains in the Columbia River Gorge, Washington State, USA. Atmos. Pollut. Res. 6 (6), 946–952. https://doi.org/10.1016/j.apr.2015.04.004.

Kim, J., Kim, H., Kweon, J., 2015. Hourly differences in air pollution on the risk of asthma exacerbation. Environ. Pollut. 203, 15–21. https://doi.org/10.1016/j.envpol.2015.03.040.

Kloog, L., Coull, B., Zanobetti, A., et al., 2012. Acute and chronic effects of particles on hospital admissions in New-England. PLoS One 7 (4), e34664.

Lane, H., Morello-Frosch, R., Marshall, J., et al., 2022. Historical redlining is associated with present-day air pollution disparities in U.S. cities. Environ. Sci. Technol. Lett. 9, 345–350. https://doi.org/10.1021/acs.estlett.1c01012.

Lelieveld, J., Haines, A., Burnett, R., et al., 2023. Air pollution deaths attributable to fossil fuels: observational and modelling study. BMJ 383, e077784. https://doi.org/10.1136/bmj-2023-077784.

Liu, J., Banerjee, S., Oroumiyeh, et al., 2022. Co-kriging with a low-cost sensor network to estimate spatial variation of brake and tire-wear metals and oxidative stress potential in Southern California. Environ. Int. 168, 107481 https://doi.org/10.1016/j.envint.2022.107481.

McDuffie, E., Martin, R., Brauer, M., 2021. A global assessment of burden of disease from exposure to major air pollution sources: synopsis of research report 210 December, 4. https://www.healtheffects.org/system/files/mcduffie-rr-210-statement.pdf.

Meyer, R., 2019. A major but little-known supporter of climate denial: freight railroads. The Atlantic. December 13, 12–13T17:43:47Z. https://www.theatlantic.com/science/archive/2019/12/freight-railroads-funded-climate-denial-decades/603559/.

Miller, K., Siscovick, D., Sheppard, L., et al., 2007. Long term exposure to air pollution and incidence of cardiovascular events in women. N. Engl. J. Med. 356, 447–458.

National Coal Council, 2019. Advancing U.S. Coal exports: an assessment of opportunities to enhance exports of U.S. Coal. https://www.nationalcoalcouncil.org/studies/2018/NCC-US-Coal-Exports-2018.pdf.

Ostro, B, 1987. Air pollution and morbidity revisited: A specification test. J. Environ. Econ. Manag. 14 (1), 87–98.

Ostro, B., Feng, W., Broadwin, R., et al., 2008. The impact of components of fine particulate matter on cardiovascular mortality in susceptible subpopulations. Occup. Environ. Med. 65 (11), 750, 6. 27.

Ostro, BD, Rothschild, S, 1989. Air pollution and acute respiratory morbidity: an observational study of multiple pollutants. J. Environ. Res. 50 (2), 238–247.

Ostro, B., Spada, N., Kuiper, H., 2023a. The impact of coal trains on PM2.5 in the San Francisco Bay Area. Air Qual Atmos Health. https://doi.org/10.1007/s11869-023-01333-0.

Ostro, B., Spada, N., Kuiper, H., 2023b. Assessment of Coal and Petroleum Coke Pollution: Air Quality, Health, and Environmental Justice. California Air Resources Board. April. https://ucdavis.app.box.com/s/sh55sgeix0r39k07zfsai1tcamux8qpw.

Parker, JD, Akinbami, LJ, Woodruff, TJ, 2009. Air pollution and childhood respiratory allergies in the United States. Environ Health Perspect 117 (1), 140–147.

Peters, A., von Klot, S., Heier, M., et al., 2004. Exposure to traffic and the onset of myocardial infarction. N. Engl. J. Med. 351, 1721–1730.

Pope III, C., Lefler, J., Ezzati, M., et al., 2019. Mortality risk and fine particulate air pollution in a large, representative cohort of US adults. Environ. Health Perspect. 127 (7), 077007.

Rabinovitch, N., Strand, M., Gelfand, E.W., 2006. Particulate levels are associated with early asthma worsening in children with persistent disease. Am. J. Respir. Crit. Care Med. 173 (10), 1098–1105.

Smithsonian Magazine, 2023. https://www.smithsonianmag.com/innovation/using-ai-to-track-air-pollution-from-open-top-coal-trains-180982846.

Tagami, P., Bridges, J., 2015. Responses and Information for City Follow-Up Questions to September 21 Informational Hearing.

Tessum, C., Paolella, D., Chambliss, S., et al., 2021. $PM_{2.5}$ polluters disproportionately and systemically affect people of color in the United States. Sci. Adv. 7 (18), eabf4491 https://doi.org/10.1126/sciadv.abf4491.

Tetreault, LF, Doucet, M, Gamache, P, et al., 2016. Childhood exposure to ambient air pollutants and the onset of asthma: an administrative cohort study in Quebec. Environ Health Perspect 124 (8), 1276–1282.

U. S. EPA, 2023b. Technical Support Document (TSD) for the 2022 PM NAAQS Reconsideration Proposal RIA: Estimating PM2.5- and Ozone-Attributable Health Benefits. U.S Environmental Protection Agency, Office of Air and Radiation Research Triangle Park. North Carolina.

U.S. Energy Information Administration, 2023. https://www.eia.gov/todayinenergy/detail.php?id=16651.

U.S. EPA, 2019. Integrated Science Assessment (ISA) for Particulate Matter (Final Report, Dec). U.S. Environmental Protection Agency, Washington, DC. https://cfpub.epa.gov/ncea/isa/recordisplay.cfm?deid=347534.

U.S, EPA, 2023a. How BenMAP-CE estimates the health and economic effects of air pollution. https://www.epa.gov/benmap/how-benmap-ce-estimates-health-and-economic-effects-air-pollution.

Vodonos, A., Awad, Y.A., Schwartz, J., 2018. The concentration-response between long-term PM2. 5 exposure and mortality; A meta-regression approach. Environ. Res. 166, 677–689 pmid:30077140.

Vohra, K., Vodonos, A., Schwartz, J., et al., 2021. Global mortality from outdoor fine particle pollution generated by fossil fuel combustion: results from GEOS-Chem. Environ. Res. 195, 110754 https://doi.org/10.1016/j.envres.2021.110754.

Wang, W., Xinye, Q., Yaguang, W., et al., 2023. Long-term exposure to ambient $PM_{2.5}$ and hospitalizations for myocardial infarction among US residents: a difference-in-differences analysis. J. Am. Heart Assoc. 12, e029428 https://doi.org/10.1161/JAHA.123.029428, 2023.

World Health Organization, 2021. WHO Global Air Quality Guidelines: Particulate Matter (PM2.5 and PM10), Ozone, Nitrogen Dioxide, Sulfur Dioxide and Carbon Monoxide. Geneva.

Wu, P.C., Cheng, T.J., Kuo, C.P., et al., 2020. Transient risk of ambient fine particulate matter on hourly cardiovascular events in Tainan City, Taiwan. PLoS One 15 (8), e0238082. https://doi.org/10.1371/journal.pone.0238082.

Yazdi, M., Wang, Y., Di, Q., et al., 2019. Schwartz J. Long-term exposure to PM2.5 and ozone and hospital admissions of Medicare participants in the Southeast USA. Environ. Int. 130 (May), 104879 https://doi.org/10.1016/j.envint.2019.05.073.

Yazdi, M., Wang, Y., Di, Q., et al., 2021. Long-term effect of exposure to lower concentrations of air pollution on mortality among Medicare participants and vulnerable subgroups. Lancet Planet. Health 5, e689–e697. https://doi.org/10.1016/S2542-5196(21)00204-7.

Yitshak-Sade, M., Lane, K.J., Fabian, M.P., et al., 2020. Race or racial segregation? Modification of the $PM_{2.5}$ and cardiovascular mortality association. PLoS One 15 (7), e0236479. https://doi.org/10.1371/journal.pone.0236479.